DataBridge High-Level Design (HLD) Document





Version 1.2.2

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Date: *11/10/2026*

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Revision History

Name	Date	Reason For Changes	Version
Alec Howard	10/30/2026	Document Creation	1.2.2
Alec Howard	11/3/2026	Document Structure/Design	1.2.2
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Saph Mabalot	11/05/2026	1.1.1 - 1.1.2	1.2.2
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1. Introduction

DataBridge is a web-based bedside documentation system developed by the 4Health Team as a senior capstone project, It is designed for nurses in rural hospitals. Nurses use a tablet to record vital signs and bedside observation for their assigned patients. The backends validates each entry, stores it, converts it into a standard HL7 version 2 (HL7v2) message, and sends it to a simulated electronic health record (EHR) receiver, which returns an acknowledgement [10]. The goal is to cut the time between bedside care and documentation while keeping the data structured and ready for exchange with other systems.

DataBridge is designed with patient privacy as a core requirement, Its security controls follow the HIPAA Security Rule [1] and related NIST and OWASP guidance [6]-[9]. The prototype only uses synthetic patient records generated with Synthea [11], so no real Protected Health Information (PHI) is accessed or stored during development, testing, or demonstrations.

1.1 Document Purpose

This High-Level Design (HLD) document turns the requirements in the DataBridge Software Requirements Specification (SRS) into a system design the team can build from. It describes the major components of the system and how they connect, including the tablet interface, the FastAPI backend, the database, and the simulated EHR receiver.

Based on the feedback from the SRS review, this document specifically defines:

- The data model, including the main database entities, their attributes, and their relationships (Section 3.2)
- The REST API endpoints the frontend uses to communicate with the backend, and which roles may call each one.
- The HL7v2 field mapping that shows how stored documentation values are placed into HL7v2 message segments and fields
- The points in the request flow where authentication, role-based access control (RBAC) checks, and audit logging occur (Sections 2.3 and 3.1.5)

This document is the reference for the detailed design and implementation. Developers should use it to build database tables, API routes, and message generation consistently. Testers should use it to write test cases for permissions, data validation, and HL7v2 output. Any changes to the requirements or priorities from the SRS are noted in Section 2.2.

1.2 Intended Audience and Reading Suggestions

This document is written for several groups. Each group can focus on the sections most relevant to them.

- Instructors and Mentor (Dr. Sakkaff, Dr. Alaqqad, Dr. Chakraborty): Should read the whole document, with special attention to the system architecture (Section 2.1), security considerations (Section 2.3), and the data model and API design (Section 3).
- 4Health Development Team: Should focus on Section 3, which covers user roles, the data model, server operations, API endpoints, and HL7v2 mapping, along with section 2.3 for required security controls
- Testers: Should focus on the RBAC permission matrix (Section 3.1.4), access control enforcement (Section 3.1.5), and the HL7v2 field mapping, since these define the expected behavior to test.
- Bedside Nurses: The primary users, Section 3.1.1 describes what nurses can do in the system and how DataBridge helps them get bedside data into the record quickly.
- Charge Nurses and Unit Leads: Section 3.1.2 describes their unit-level view, which helps them keep track of patient's information organized and up to date.
- Physicians: Do not use DataBridge directly, but benefit from timely, structured documentation delivered to the EHR. Section 2.2 summarizes what information the system provides.
- Hospital Administration and IT Staff: Should read Section 3.1.3 for administrator responsibilities, Section 2.3 for security and HIPAA considerations, and Section 3.3 for server operations, backup and recovery,
- Patients: Are not direct users, but benefit from accurate records and strong privacy protection. Section 2.3 explains how patient information is protected.

2. Databridge Overview

DataBridge is a browser-based clinical documentation system designed to modernize bedside charting and demonstrate interoperability between a point-of-care application and an electronic health record

(EHR) system. The system allows authorized healthcare personnel to access patient information, enter and review clinical documentation, validate submitted information, and transmit supported clinical data using the HL7 Version 2 (HL7v2) messaging standard.

DataBridge uses a modular client-server architecture. The browser-based frontend provides the user interface for Nurses, Charge Nurses, and Administrators. The frontend communicates with the backend through HTTPS-based REST API requests. The backend is responsible for authentication, multi-factor authentication (MFA), role-based access control (RBAC), input validation, documentation processing, HL7v2 message generation, and audit logging.

The backend uses a SQLite database to persist application information. The primary data entities include User, Patient, Documentation, HL7v2Message, and AuditLog. Synthetic patient information generated through Synthea is imported using a Python data-import script. This allows the project to demonstrate patient-data workflows without relying on real patient information.

For interoperability testing, DataBridge communicates with an external simulated EHR receiver. The backend sends HL7v2 messages to the receiver and processes the receiver's acknowledgment or rejection response. This allows the team to demonstrate the message exchange process without connecting to a production hospital EHR system.

The modular architecture separates the user interface, application logic, data persistence, and external communication responsibilities. This separation makes the system easier to maintain, test, and extend.

It also provides defined locations for security checks, validation, and audit logging throughout the application's request-processing workflow.

2.1 System Architecture

2.1.1 Architecture Diagram

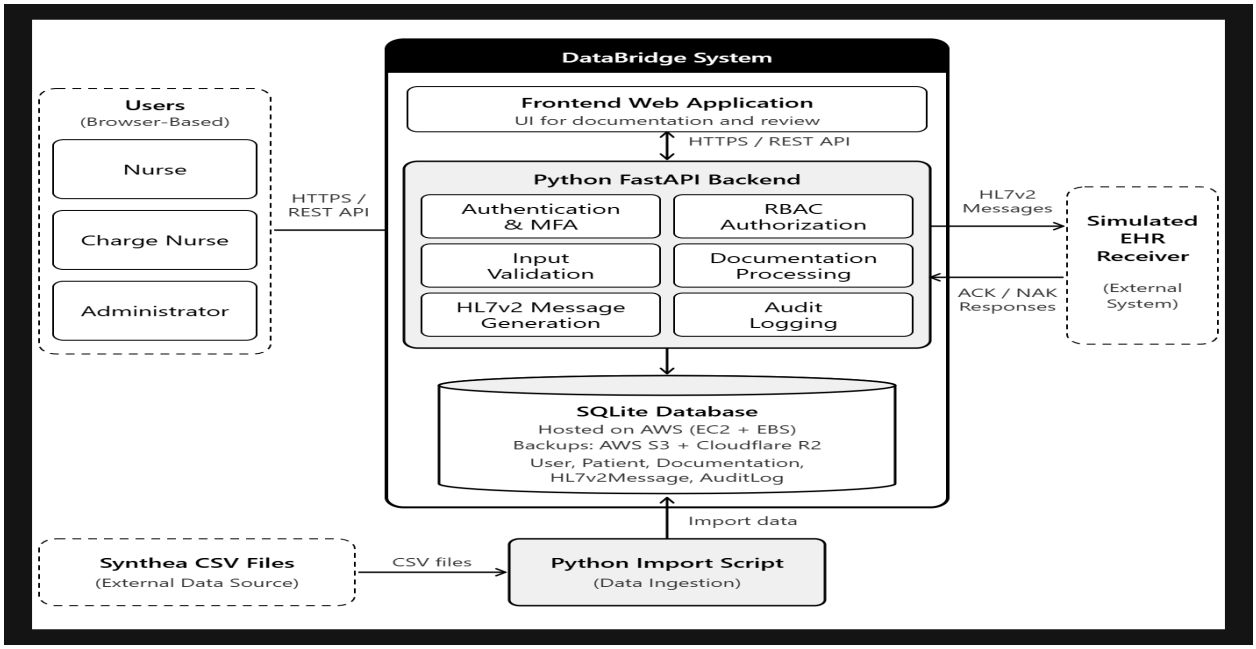


Figure 2.1.1: Architecture Diagram

Figure 2.1.1 illustrates the high-level architecture of DataBridge. It identifies the browser-based users, frontend application, FastAPI backend, database, synthetic-data import process, and

external simulated EHR receiver. The diagram also identifies the major backend modules and the communication paths between system components.

The DataBridge system boundary separates application-owned components from external systems.

Synthea is represented as an external source of synthetic patient data, while the simulated EHR receiver is represented as an external destination for HL7v2 messages. Users access the system through the browser-based frontend, and all application requests are processed through the backend API.

2.1.2 HIPO Diagram

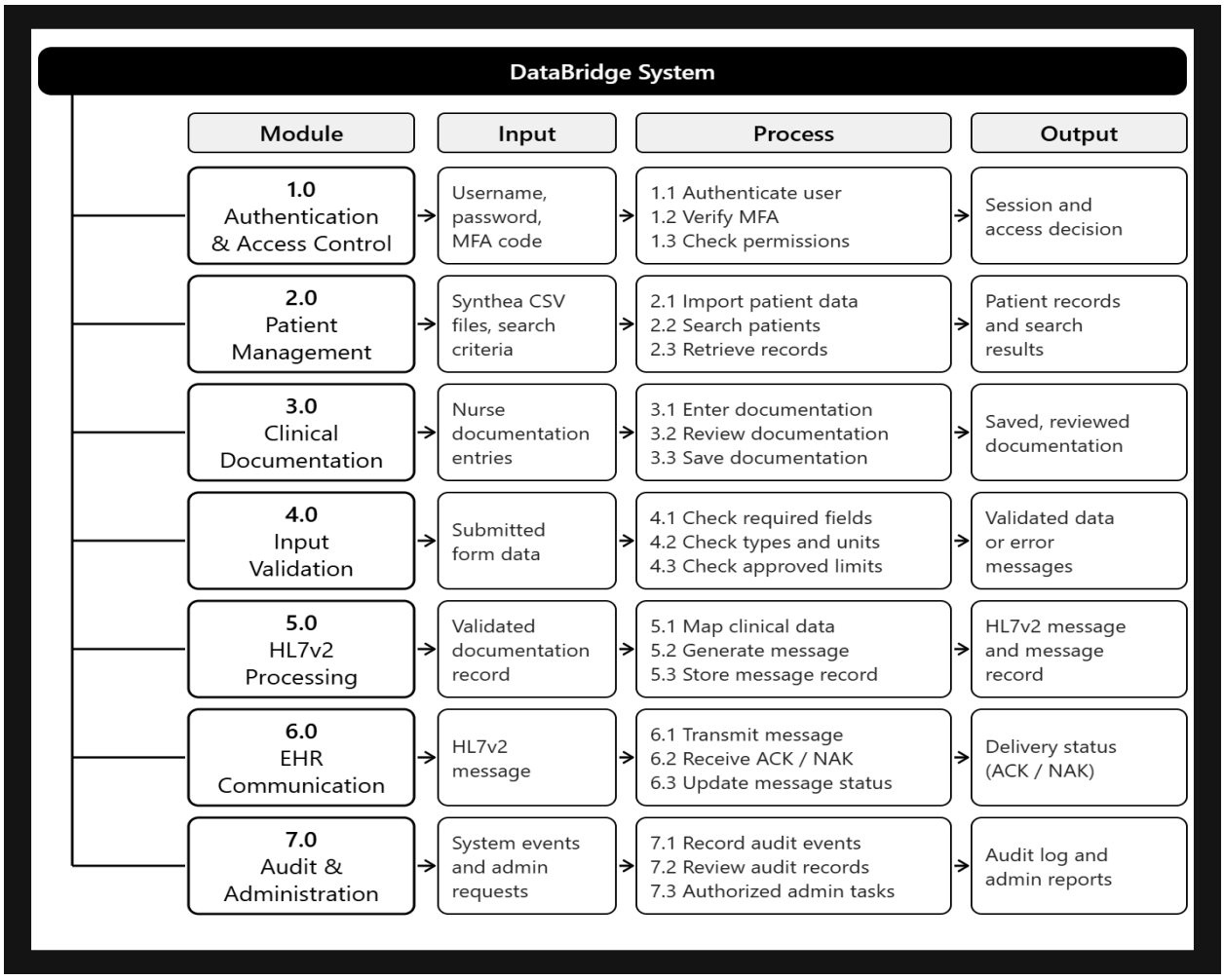


Figure 2.1.2: HIPO Diagram

The Hierarchy plus Input-Process-Output (HIPO) model describes the major functional modules of DataBridge and identifies the information each module receives, the operations it performs, and the resulting output. The system is divided into authentication and access control, patient

management, clinical documentation, validation, HL7V2 processing, EHR communication, and audit management.

This decomposition illustrates how the application's major responsibilities are separated while remaining part of one integrated system.

2.1.3 Data Flow Diagram

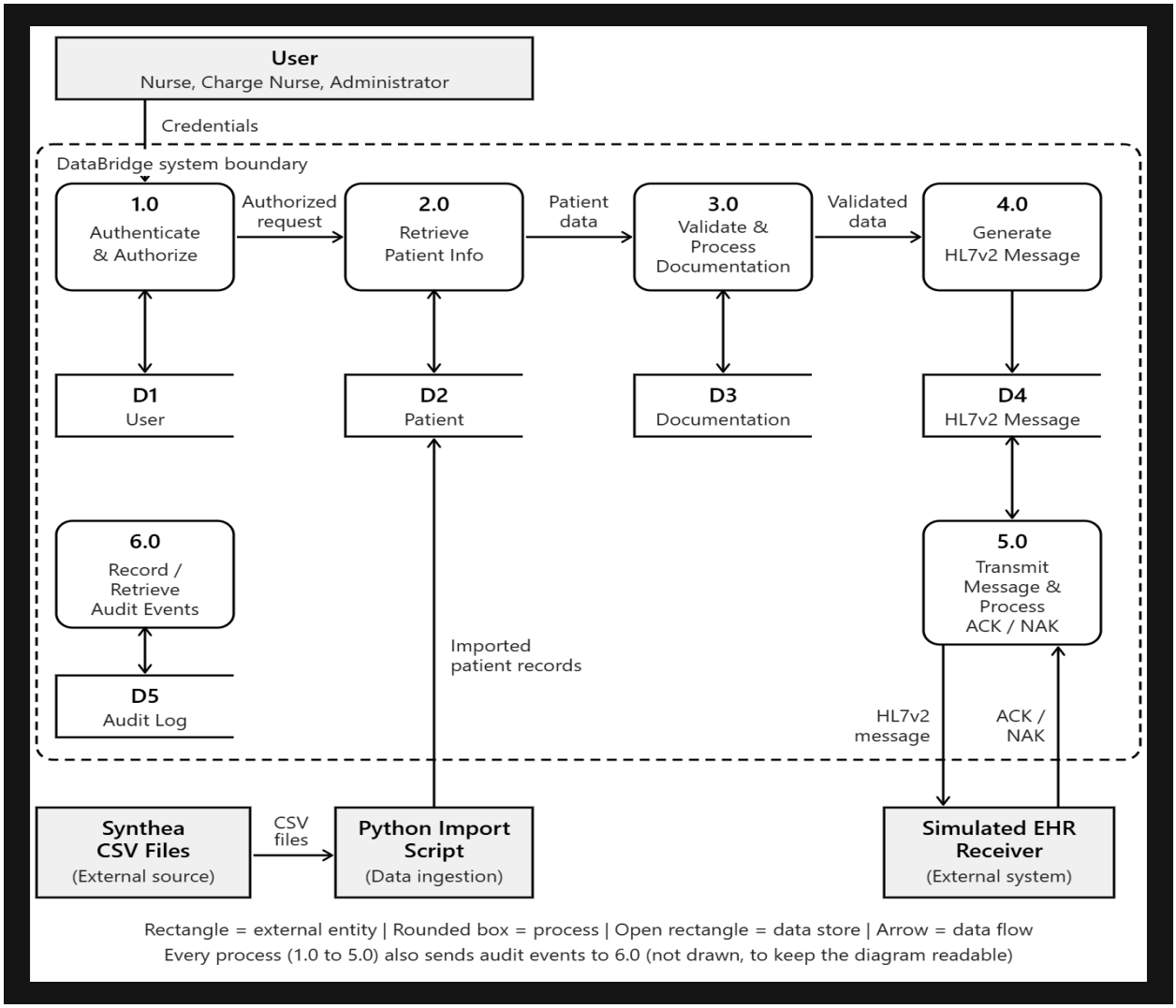


Figure 2.1.3: Data Flow Diagram

The Data Flow Diagram (DFD) illustrates how information moves through DataBridge. It identifies the external entities that interact with the application, the processes that transform or validate information, and the data stores used to persist application records.

A user submits a request through the browser-based frontend. The backend authenticates the user and checks the user's role before allowing access to the requested functionality. When a user

submits clinical documentation, the backend validates the input, associates the documentation with the appropriate patient and user records, and stores the resulting information in the database.

When HL7v2 transmission is required, the system generates a message from the validated documentation and sends it to the simulated EHR receiver. The receiver returns an acknowledgment or rejection response. DataBridge processes that response, updates the associated message record, and records relevant events in the audit log.

Synthetic patient data follows a separate ingestion process. Synthea CSV files are supplied to the Python import script, which processes the records and inserts the resulting patient data into the database.

2.1.4 High-Level Sequence Diagram

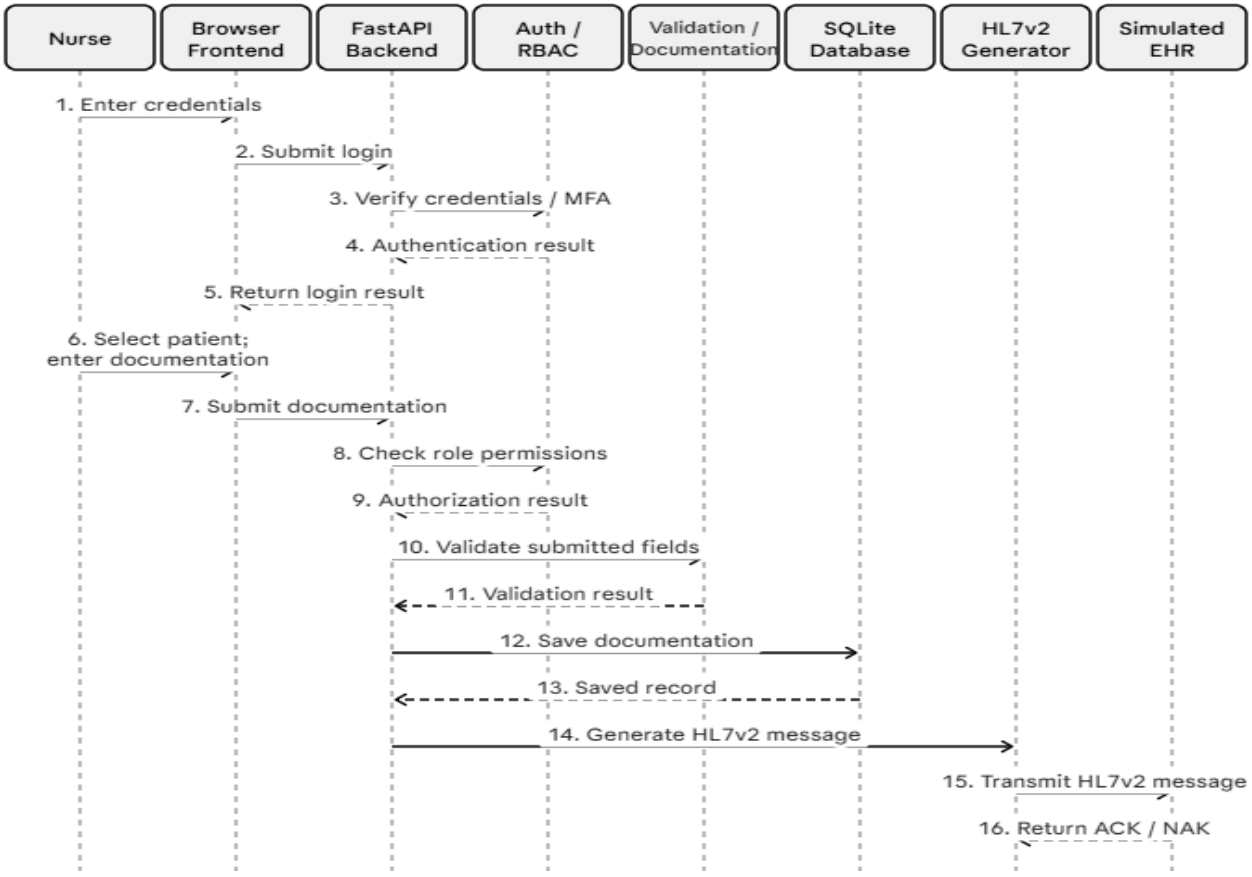


Table 2.1.4: High-Level Sequence Diagram

The high-level sequence diagram illustrates the order of interactions that occur when a Nurse submits clinical documentation for a patient. The sequence begins when the user accesses

DataBridge through the browser and authenticates. If MFA is enabled for the account, the required verification step must also be completed before access is granted.

After authentication, the backend checks the user's permissions through RBAC. The user then selects a patient and submits documentation. The backend validates the submitted information before saving it to the database. If the documentation is eligible for transmission, the backend generates the corresponding HL7v2 message and sends it to the simulated EHR receiver.

The receiver returns an acknowledgment or rejection response. DataBridge processes the response, updates the message status, and records the relevant activity in the audit log. The frontend then displays the resulting submission or transmission status to the user.

The sequence diagram should also represent alternate outcomes, including failed authentication, insufficient permissions, invalid documentation, and unsuccessful HL7v2 transmission.

2.2 Summary of Product Functions

DataBridge provides a set of related functions that support bedside documentation, patient information management, security, and HL7v2 interoperability.

User Authentication and Access Control: DataBridge authenticates users before granting access to protected'application functionality. MFA is included as a security requirement, subject to the team's finalized implementation. RBAC determines which operations are available to Nurses, Charge Nurses, and Administrators.

Patient Information Management: The system provides authorized users with access to synthetic patient information stored in the database. Patient records are initially populated using Synthea-generated CSV files processed through a Python import script.

Clinical Documentation: Authorized users can enter and review bedside clinical documentation associated with a patient. Documentation is processed by the backend and stored in the database for later retrieval and interoperability processing.

Input Validation: The backend validates submitted documentation before it is stored or used to generate an HL7v2 message. Validation includes required fields, data types, units, and approved value constraints. The final validation rules must be consistent with the requirements in DBR-DOC-05 and DBR-BVAL-02.

HL7v2 Message Generation: DataBridge converts supported clinical documentation into an HL7v2 message using the field mappings and message structure established by the project. The generated message is associated with its source documentation and stored in the database.

EHR Communication: The system sends generated HL7v2 messages to an external simulated EHR receiver. It processes the receiver's acknowledgment or rejection response and records the resulting transmission status.

Audit Logging: DataBridge records relevant application events, including authentication activity, documentation operations, and HL7v2 transmission events. Audit records provide traceability for actions performed within the system.

Charge Nurse Functions: The Charge Nurse role should support the additional functions established in the revised requirements, including the patient census and alert-related capabilities specified by the team. Access to these functions must be enforced through backend authorization checks.

Administrative Functions: Administrators are provided with the application-management capabilities defined in the requirements. Administrative access must still pass through the normal authentication and authorization process.

Optical Character Recognition (OCR): If retained in the final project scope, OCR supports the extraction of information from supported documents or images. Its library, processing workflow, and validation behavior must be specified once the team finalizes the implementation.

The final HLD should reflect any approved changes to the SRS. In particular, MFA and Charge Nurse census and alert functionality should be represented consistently in the requirements, architecture, diagrams, and implementation. Any requirements that remain undecided should be identified as pending design decisions rather than described as completed functionality.

2.3 Security Considerations

DataBridge architecture has been built with the aim of safeguarding patient data using various layers of security that include encryption, user authentication, access control, secure transmission, and audit trail. Considering the fact that the product is meant to facilitate the creation of medical documentation processes, its security model is based on the confidentiality, integrity, and availability of patient data. Security requirements for the product are informed by HIPAA guidelines, NIST publications, and application security standards.

2.3.1 Data Encryption and Secure Communication

Information security is important during the transmission of the data among the following components of DataBridge: tablet interface, backend services, database, and simulated EHR recipient. It is recommended to secure communications with the help of HTTPS using TLS to ensure that third parties cannot intercept or read information being transmitted. TLS 1.2 or higher versions should be used, and the TLS 1.3 version is preferable.

Sensitive data kept in the system should be encrypted or secured in any other way documented to avoid possible unauthorized access.

2.3.2 Authentication and Password Security

DataBridge will require user authentication before gaining access to the DataBridge. User accounts are meant to have individual user names and passwords that will be used for identification purposes. Passwords should not be stored in clear text. A strong, adaptive password hashing algorithm such as Argon2id should be implemented.

Multifactor authentication (MFA) is part of the security considerations to offer additional identity verification methods. According to updated documentation, physical security keys are among the options that can be used as MFA mechanisms especially for users who have higher privilege levels.

2.3.3 Role-Based Access Control (RBAC)

DataBridge uses Role-Based Access Control (RBAC) that restricts access to functionalities of the system and patient data depending on the user's responsibilities. There are three primary user roles as follows:

> **Nurse:** *Can access patient data assigned to her/him and perform the bedside documentation.*
> **Charge Nurse:** *Can access patient data at the unit level and control nursing documentation operations.*
> **Administrator:** *Controls users, grants permissions and is capable of accessing audit logs of the system.*

Permissions must follow the least privilege approach meaning that users are granted minimum necessary permissions to perform assigned duties. User authentication and authorization must occur prior to accessing system resources. Users are unable to bypass permission limits imposed by roles through use of backend functions.

2.3.4 Audit Logging and Accountability

The DataBridge uses audit logging for accountability and tracking of significant events within the system. These events must specify who performed the action and what timestamp it occurred at. Logs should be secured from tampering, alterations, or any form of unauthorized access.

Administrative users have the responsibility to examine the audit logs as needed. Audit logs will not be visible on the tablet display that nurses use. The simulated EHR receiver has the responsibility to send back acknowledgments and does not write audit logs.

2.3.5 Secure Document Sharing

DataBridge utilizes a method of document exchange that is secure since it uses authentication and temporary links. The links are supposed to expire after a specified amount of time or number of uses in order to minimize the chance of any access without authorization. Any user accessing a protected document needs to authenticate themselves first.

2.3.6 HIPAA Regulatory Considerations

DataBridge security design seeks to meet HIPAA Security Rule standards that set out the appropriate safeguards for securing electronic Protected Health Information (ePHI). The design considers the following three major categories of safeguards:

- **Administrative Safeguards:** *These include security policies, user responsibilities, access, risk assessment, and workforce security procedures.*
- **Physical Safeguards:** *This involves the protection of the devices, workstations, tablets, and facilities from where access to or storage of the system's information takes place.*
- **Technical Safeguards:** *These include access control, user authentication, audit controls, data integrity, and transmission security for electronic information.*

Applicable HIPAA requirements can be found in 45 CFR Part 164, Subpart C. Specifically, sections 164.308, 164.310, and 164.312 contain requirements for administrative, physical, and technical safeguards respectively. Encryption is an addressable implementation specification under the HIPAA Security Rule, meaning that an organization has to assess the appropriateness of encryption and document its implementation decision.

The current DataBridge capstone prototype utilizes the synthetic patient information created using Synthea and a simulated EHR receiver. This prototype does not handle any live patient record or production ePHI. Therefore DataBridge is to be considered to be HIPAA-aligned in its design considerations.

2.3.7 Engineering Standards and Regulatory Codes

The following regulations, standards, and technical guidelines provide the foundation for DataBridge:

Standard / Code	Reference Number(#)	Application to DataBridge

HIPAA Security Rule	45 CFR Part 164, Subpart C	Establishes security requirements for protecting electronic patient information.
Administrative Safeguards	45 CFR §164.308	Guides risk management, user responsibilities, and workforce access procedures.
Physical Safeguards	45 CFR §164.310	Guides protection of tablets, workstations, devices, and physical access to the system.
Technical Safeguards	45 CFR §164.312	Supports authentication, access control, audit logging, data integrity, and secure transmission.
Security Documentation	45 CFR §164.316	Establishes requirements for maintaining security policies, procedures, and documentation.
NIST Digital Identity Guidelines	NIST SP 800-63B-4	Provides guidance for password security, authentication, and multifactor authentication.
NIST Transport Layer Security Guidelines	NIST SP 800-52 Rev. 2	Provides guidance for configuring secure TLS communications between system components.
OWASP Application Security Verification Standard	OWASP ASVS 5.0	Provides security requirements for application development, authentication, authorization, and data protection.
OWASP Password Storage Guidelines	Password Storage Cheat Sheet	Recommends secure password hashing and protection of stored credentials.
Health Level Seven Messaging Standard	HL7 Version 2 (HL7v2)	Defines the healthcare messaging format used for communication

Table 2.3: Standards and Codes

2.3.8 Security Design Summary

The DataBridge security architecture combines authentication, RBAC, encryption, audit logging, and temporary document-sharing links to reduce the risk of unauthorized access and exposure of sensitive information. These controls support the system's intended healthcare workflow while providing a foundation for future security improvements. All security features should be implemented and evaluated according to the applicable regulatory requirements and engineering standards identified in this section.

3. Design Details

3.1 Users and Their Roles

The user types supported by DataBridge are Nurse's, Charge Nurse's, and Administrators. These different user types have been allotted permissions depending upon their respective roles. Role-Based Access Control (RBAC) mechanism has been used in order to decide whether the authenticated user has any permission for certain function(s) and/or patient data.

Users are required to authenticate themselves in order to access the protected functionality of the system. Post authentication, the backend verifies the type of role associated with the user and checks his/her permissions.

3.1.1 Nurse (Primary User)

The Nurse is the main user of the DataBridge and is the one who has to document patient data while on the job. The nurse accesses the system via a web-enabled tablet interface that allows for fast and efficient documentation while on shift.

The Nurse permissions include:

- Authenticate via a designated user account.
- Access patients assigned to him or her from the active patient census.
- Document patient vital signs and bedside observations.
- Access patient data pertinent to his or her authorization.
- Get confirmation of successful documentation submission.

Nurses can only access data on their own patients and do not have access to any administrative functionalities or audit log records.

3.1.2 Charge Nurse (Secondary User)

Charge Nurse has wider patient management duties than a single Bedside Nurse. It allows access to the patient census at the unit level.

The following permissions apply to the Charge Nurse:

- Log on using the assigned user id.
- Access the patient census at the unit level.
- Access patient records at the unit level within the authorized scope.
- Perform documentation duties at the bedside level.
- Monitor patients and coordinate nursing work.

The Charge Nurse has more visibility to patients than the nurse but does not have unrestricted administrator or user management rights.

3.1.3 Administrator

Administrator is tasked with system access management and security tasks within the software system. This user category enjoys the highest form of administrative privileges in the application and is not designed for use in documenting patients at the bedside.

Administrator privileges allow for:

- Authenticating into the system with the provided administrative account.
- Creating, maintaining and managing user accounts.
- Assigning and managing user roles and privileges.
- Monitoring audit logs and user activities in the system.
- Carrying out other approved system configuration tasks.

Administrative privilege is restricted to designated users.

3.1.4 Role-Based Permission Matrix

Table 3.1 identifies the primary permissions associated with each DataBridge user role.

System Function / Permission	Nurse	Charge Nurse	Administrator
Authenticate / Log In	✓	✓	✓
View Assigned Patients	✓	✓	—
View Full Unit Census	—	✓	✓
Enter Patient Vital Signs	✓	✓	—
Add Bedside Observation Notes	✓	✓	—
Submit Clinical Documentation	✓	✓	—
Manage User Accounts	—	—	✓
Assign User Roles and Permissions	—	—	✓
Review System Audit Logs	—	—	✓
Administrative System Functions	—	—	✓

Table 3.1: RBAC Permissions

3.1.5 Access Control Enforcement

Role-based permission control is carried out using DataBridge backend API as opposed to just interface limitations. Any request which needs authorization is to be authenticated based on the user's identity and role.

Steps in the authorization process include the following:

1. The user opens the DataBridge login page.
2. The user fills in the correct authentication credentials.
3. The identity of the user is verified.
4. The role assigned to the user is verified.
5. The backend checks if the requested activity is permissible for that particular role.
6. Access to the activity is provided if the authorization process is successful or refused otherwise.
7. User activity is logged.

This process has been put in place in order to ensure that no user uses the backend access to override the permissions available.

3.2 Entity Relationships

The DataBridge database was created for the purpose of storing and organizing information about system users, patients, documentation, HL7v2 messages, and audits. These components are interconnected in order to implement the process of bedside documentation, user management, and communication with the simulated EHR receiver.

Relationships between entities are used for ensuring data integrity and linking users, patient data, and documentation.

3.2.1 Main Database Entities

The primary database entities are described below:

1. User

The User entity holds data for users authorized to use the DataBridge software. The User entity includes the role of the user which is either NURSE, CHARGE_NURSE, or ADMIN depending on the permission level. The User entity also has the account status of the user.

2. Patient

Patient holds patient information to be used for bedside documentation workflow. Patient records consist of a unique identifier, patient information obtained from Synthea, a bed or location assignment, and the patient status. The statuses of the patient include: STABLE, ALERT, and MONITORING.

3. Documentation

Documentation contains the patient's clinical data that is documented via the bedside interface. The data includes the vital signs of the patient, the observations made, and the date of submission. The Documentation object will be tied to the particular patient and the submitter.

4. HL7v2Message

The HL7v2Message entity manages the structured healthcare messages generated from validated documentation. It stores message content, creation time, and transmission status. Message status values include PENDING, SENT, ACKNOWLEDGED, ERROR, and REJECTED.

5. AuditLog

The AuditLog entity records relevant system activity for accountability and security monitoring. Each audit record identifies the associated user and records the activity and timestamp. Access to audit records is restricted to authorized users, and unauthorized modification is prevented.

3.2.2 Entity Relationships

The primary database relationships are described as follows:

- User to Documentation: One User can submit multiple Documentation records. Each Documentation record is associated with the User who submitted it.

- Patient to Documentation: One Patient can have multiple Documentation records over time. Each Documentation record belongs to one Patient.

- Documentation to HL7v2Message: A Documentation record can generate an HL7v2Message for transmission to the simulated EHR receiver. This relationship connects submitted information to its corresponding interoperability message.

- User to AuditLog: One User can be associated with Audit records. Each Audit record identifies the User for the recorded activity.

- These relationships allow DataBridge to maintain connections between Documentation, the Users who submit it, the Patients associated with it and the Messages generated for the simulated EHR.

3.2.3 Entity Relationship Diagram

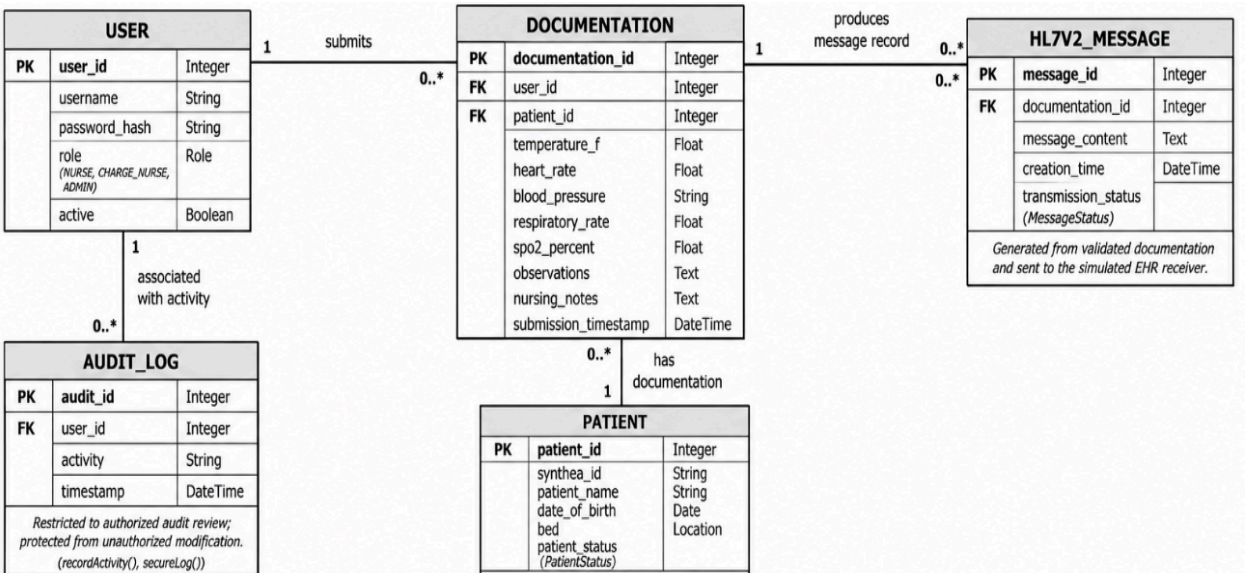


Figure 3.2.3: ERD

3.3 DataBridge Server Operations

3.3.1 [DBSO] Standard Operations:

DataBridge enables authorized users such as nurses, charge nurses and administrators to gain access to the system via a web browser. The login process enables authentication while Role-Based Access Control (RBAC) dictates which functions are enabled depending on each user's privileges. The clinical notes and patient data are submitted to the backend server that will validate and process the entries before transmitting HL7v2 messages to the EHR receiver. The system actions will be logged in the audit log.

3.3.2 [DBSO] Offline Actions:

For normal operation, DataBridge requires a connection to the network as it interacts with both the backend server and the EHR receiver. In case there is no connection, users might not be able to submit the clinical notes. Resubmitting should be avoided until the system is up and running again and the status of the last submission is confirmed.

3.3.3 [DBSO] Shutdown:

In order to shutdown DataBridge, users must stop the backend server using the relevant development environment tools. Users must finish pending submissions before shutting down the program. All databases and project files should be kept intact when the program is shut down.

3.3.4 [DBSO] Restart:

For DataBridge to be restarted, the 4Health Team initiates the backend server and ensures that the database connection and other application components are working well. The web interface can then be accessed via a browser and tested by logging in and submitting data.

3.3.5 [DBSO] Backup:

Project Database and any important configuration files need to be backed up regularly. These backups need to be kept secure and any patient information, credentials, or encryption keys need to be kept away from any unauthorized parties. The regularity of backups depends on the needs of the project and the volume of the data that needs to be kept.

3.3.6 [DBSO] Recovery:

In case of an application failure for DataBridge, the 4Health Team looks into the server outputs, database connections, and the audit logs to figure out the problem. The backend or the affected components can be restarted once the problem has been sorted out. In case of database corruption or data loss, the latest backup that has been confirmed can be restored. Post-recovery, the 4Health Team will have to check that authentication, permission levels, clinical note processing, HL7v2 messages and EHR acknowledgements are all working well.

3.4.1 Class Diagram:

Figure 3.4.1 presents the UML class diagram for DataBridge. It models the persistent domain entities, the enumerated types that constrain their values, and the external receiver with which the backend exchanges HL7v2 messages. The classes, attributes, and multiplicities are consistent with the Entity Relationship Diagram (Figure 3.2.3), the UML model in the SRS (Appendix C), and the data model in Appendix A.1.

3.4.1 Class Diagram

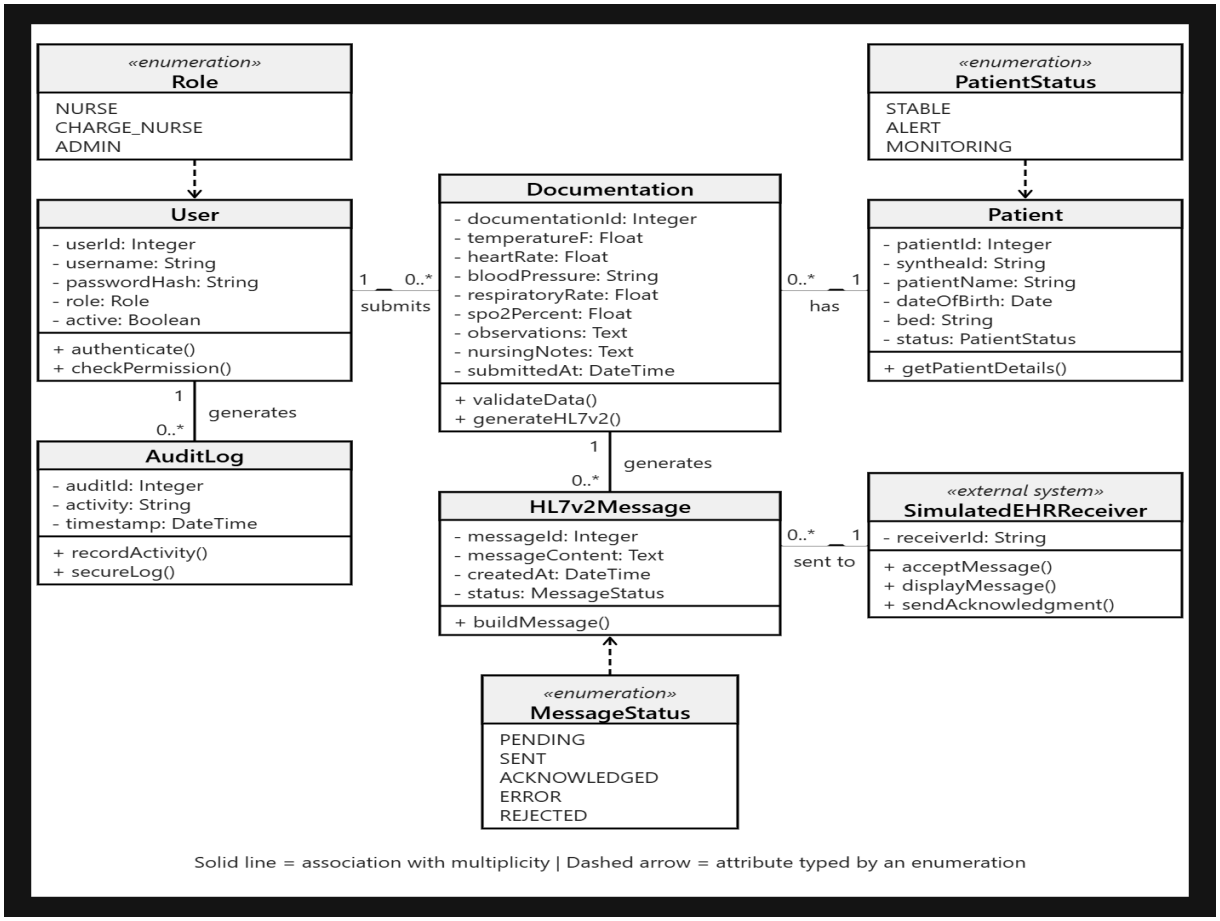


Figure 3.4.1: Class Diagram

3.5.1 Dependencies and Assumptions

Dependencies: The external tools, services and data we use/system relies on to be dev'ed and function

- Backend Framework: The system will depend on Python and the FastAPI framework to handle backend validation, authentication, API requests, and communication between the frontend and simulated EHR receiver.
- Frontend Technologies: The system will depend on React and/or the standard HTML, CSS and Java to provide the browser-based user interface. The frontend will support tablet and desktop screen sizing.
- Simulated EHR Receiver: The system will depend on separately developed simulated EHR receiver in order to accept and process incoming HL7v2 messages, then display received patient information for verification.
- Authentication and Security: The system will depend on an authentication mechanism, role-based access control, and encryption libraries to manage user permissions and protect sensitive information.
- Synthetic Patient Data: The system will depend on synthetic patient data that is generated through tools such as Synthea in order to populate the patient census and give a realistic testing scenario without using real PHI.

Assumptions: Conditions/expects considered 'true' during dev and implementation which influence the system design

- User Environment: It will be assumed that the nurses and all other authorized users will access DataBridge through a modern web browser on a tablet, or desktop.
- User Accounts and Roles: It will be assumed that each user will have an established account with an assigned role before accessing the system.
- Patient Assignment: It will be assumed that nurses will *only* be able to access the patients assigned to them, while charge nurses will have access to the full unit's census. With this, Administrators will have access to the system management and audit information.
- HL7v2 Compatibility: It is assumed that the simulated EHR receiver will accept and correctly interpret HL7v2 messages which are generated by DataBridge.
- Synthetic Data Only: It will be assumed and declared that all patient information used throughout the development, testing, and demonstrations are purely synthetic, and that no real patient information OR PHI will be accessed and/or stored.
- Project Scope: It will be assumed that the system will *only* support bedside vitals, observation notes, file organization, and HL7v2 messaging. Features that involve medication administration, medical diagnosis, prescriptions, scheduling, billing, and full EHR functionality are outside of the scope of this project.

4. Technical Data Appendix

A.1 Data Model

The DataBridge data model defines how system information is organized, stored, and connected. The primary entities include:

Entity	Description
User	Stores usernames, password hashes, assigned roles, and account status.
Patient	Stores synthetic patient information imported from Synthea.
Documentation	Stores patient vital signs, observations, nursing notes, and submission timestamps.
HL7v2Message	Stores generated HL7v2 messages and their transmission status.
AuditLog	Records user activities and timestamps for security monitoring.

Table A.1: Data Model

The primary relationships are:

- One User can submit multiple Documentation records.
- One Patient can have multiple Documentation records.
- One Documentation record can generate multiple HL7v2Message records.
- One User can be associated with multiple AuditLog records.

A.2 REST API Endpoints

DataBridge uses FastAPI to provide communication between the browser interface, backend services, and database. The following table identifies the primary API endpoints.

Method	Endpoint	Purpose
POST	/api/auth/login	Authenticates users.
POST	/api/auth/logout	Ends an authenticated session.
GET	/api/patients	Retrieves the authorized patient census.
GET	/api/patients/{id}	Retrieves selected patient information.
POST	/api/documentation	Submits clinical documentation.

GET	/api/documentation/{id}	Retrieves a documentation record.
GET	/api/hl7/messages/{id}	Retrieves HL7v2 message status.
GET	/api/admin/audit-logs	Allows administrators to review audit records.

Table A.2: API Endpoints

Protected endpoints require authentication and backend authorization before processing requests. These are the proposed endpoint definitions and should match the final implemented API routes.

A.3 HL7v2 Field Mapping

DataBridge converts validated clinical documentation into HL7v2 messages before transmitting the information to the simulated EHR receiver.

DataBridge Information	HL7v2 Segment	Field
Message Identifier	MS	MS-10
Message Type	MS	MS-9
Message Timestamp	MS	MS-7
Patient Identifier	PID	PID-3
Patient Name	PID	PID-5
Date of Birth	PID	PID-7
Patient Bed Location	PV1	PV1-3
Vital Signs	OBR	OBR-3 and OBR-5
Measurement Units	OBR	OBR-6
Observation Timestamp	OBR	OBR-14

Table A.3: HL7v2 Field Mapping

OBR stands for Observation/Result. It is an HL7v2 segment used to communicate clinical measurements and observations, which makes it useful for DataBridge.

The simulated EHR receiver returns an acknowledgment after processing the message:

- **AA – Accepted:** Message successfully accepted.
- **AE – Error:** Message processing encountered an error.
- **AR – Rejected:** Message was rejected.

DataBridge updates the corresponding HL7v2 message status based on the acknowledgment received.

i.e. Imagine a nurse enters a patient's vital signs into DataBridge.

- MSH: Identifies the message being sent.
- PID: Identifies which patient the information belongs to.
- PV1: Identifies the patient's visit or hospital location.
- OBX: Contains the actual vital signs recorded by the nurse.

This appendix describes the data model, REST API architecture, and HL7v2 messaging framework that has been implemented in the DataBridge system. It details how the backend handles patient data management, documentation processing, authentication and role-based access control, as well as interaction with the simulation of an EHR receiver.

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